

Seminar 3

Stored Procedures, Global Variables, Dynamic Execution, the OUTPUT clause. Cursors in SQL SERVER

I Stored Procedures

- contains a group of SQL statements

```
CREATE PROCEDURE (sau PROC) usp-Name
```

```
AS  
BEGIN
```

```
-----
```

```
END
```

```
GO -- batch separator
```

```
EXEC usp-Name ----  
usp-Name --
```

```
EXECUTE
```

dacă declar variabile
înainte de GO, după GO nu o să
mai fii valabile !

ex. CREATE PROC usp-HotelName

```
AS  
SELECT hname  
FROM Hotel  
GO
```

```
EXEC usp-HotelName
```

← param

```
ALTER proc usp-HotelNames (@no-stars DECIMAL(0,2))  
AS
```

```
    SELECT hname  
    FROM Hotel  
    WHERE STARS = @no-stars  
GO  
EXEC usp-HotelNames 5.0
```

-- Counting hotels

```
CREATE PROC usp-H (@no-stars DECIMAL(10,2) = 3.0, @NoHotels  
INT OUTPUT)
```

```
AS  
    SELECT @NoHotels = count(*)  
    FROM Hotel  
    WHERE stars = NoStars
```

```
GO  
DECLARE @ NoH INT = 0  
SET @ NoH = 0
```

we can't create
our global var.

II Global Variables

eg. @@ ERROR

@@ IDENTITY

@@ ROWCOUNT

@@ SERVERNAME

@@ SPID (process id of the current session)

@@ VERSION

III Dynamic Execution

```
EXEC ('select * from Hotel')
```

eg. declare @id ngl varchar(1000 = '27')

declare @select_sql varchar(max)

set @select_sql = 'select * from Hotel where hid = 27' + @id + '_sql'

exec (@select_sql)

IV Output Clause

- access inserted, deleted

- UPDATE → inserted + deleted

- INSERT → inserted

- DELETE → deleted

eg. declare @LogTable Table (hid INT, hname VARCHAR(200), no_rooms_old INT, no_rooms_new INT, username VARCHAR(100), dateofupdate DATETIME)

UPDATE Hotel

SET no_rooms = 100

OUTPUT inserted - hid, inserted hname, deleted - no_rooms, inserted no_rooms

INTO @LogTable

WHERE hid = 27

select * from @LogTable

(for delete and insert we write them instead of update)

V Cursors

- inefficient

DECLARE CURSOR

↳ specify the result set, using a select stmt

↓
it won't get executed

OPEN

↳ the associated select stmt will get executed

FETCH

FIRST

LAST

NEXT

PRIOR

ABSOLUTE n

↳ int

$n > 0$ - n^{th} row starting from the 1st row

$n < 0$ - n^{th} row before the last row

$n = 0$ - no rows

RELATIVE n , $n - INT$

$n > 0$ - the n^{th} row after the current row

$n < 0$ - n^{th} row before the current row

$n = 0$ - cursor is set on the current row

UPDATE / INSERT

CLOSE - some resources are freed eg. the result set.

DEALLOCATE

